Plastics can be grouped in two categories: thermoplastics and thermosets. While thermoplastics can be remoulded 2-3 times into any shape after being heated and are therefore easy to recycle, thermosets can be moulded only once. Here are the different kinds of plastics, their uses and recyclability

THERMOPLASTICS



POLYVINYL CHLORIDE (PVC)

Used in: Pipes, weather-resistant clothes, medical devices, toys, kitchen cling wraps, wires, shoe soles

Recyclability: Often not recyclable



LOW DENSITY POLYETHYLENE (LDPE)

Used in: Plastic bags, juice and milk pouches, electrical components

Recyclability: Not recyclable



POLYPROPYLENE (PP)

Used in: Bottle caps, plastic cutlery, takeaway food containers, raffia bags, buckets, mugs, tubs, chairs, automobile parts

Recyclability: Often not recyclable



POLYSTYRENE (PS)

Used in: Home appliances like AC, TV, water purifier

Recyclability: Not recyclable

Source: Recykal

THERMOSETS

EPOXY RESIN, PHENOLIC RESIN, BAKELITE, MELAMINE, VULCANISED RUBBER

Used in: Car parts, mattresses, circuit boards, jewellery

Recyclability: Not recyclable



Indian cities, found Delhi and Gwalior the least compliant, with 88 and 84 per cent points, respectively, still using banned SUPs, followed by Guwahati (77 per cent), Mumbai (71 per cent) and Bengaluru (55 per cent). Depressingly, carry bags less than 120 microns were commonly available, and other banned products, including thermocol decorations (74 per cent), balloons and plastic stick earbuds (60 per cent each), continued to be sold in all cities.

One of the top reasons for the lack of enforcement, says Swati Singh Sambyal, senior circular economy expert at Norwegian non-profit GRID-Arendal, is the lack of rigour in state pollution control boards and local governments, including municipalities. This, she explains, is compounded by insufficient capacities of the enforcers and manpower. "Additionally, the industry for plastic alternatives remains relatively small and less specialised." SUP alternatives include bamboo cutlery, areca palm leaf tableware or paper plates, paper cups with plant-based lining, beeswax food wraps instead of cling films etc. But these remain expensive and inaccessible across income groups, which is why mainstream adoption of SUP alternatives stays low.

Siddharth Ghanshyam Singh, programme manager, municipal solid waste, CSE, sees the problem of plastic pollution as more than waste management. Waste management principles adhere to an order of priority: reduce, reuse, recycle, recover. We seem to falter at the very first step—globally, the production of plastic is only increasing. As a result, a 2023 report by Swiss research consultancy Earth Action reveals that, even with improvement in waste management capacity, global plastic pollution will triple by 2040. Wilma Rodrigues, founder and chief transformation officer of Bengaluru-based social enterprise Saahas Zero Waste, says, "We can't continue producing plastic without the proper means to manage the waste." Singh has an issue with the material itself. "It poses a challenge at every stage of its life cycle, from production to disposal," he says. To prevent plastic pollution, each stakeholder must play a part. From the big petrochemical companies like Indian Oil Corporation, GAIL, BPCL and Reliance that supply the polymers, to the manufacturers—mostly MSMEs—that fashion the polymers into products catering to their vendors, to the brands, retailers and consumers who ultimately distribute and use these products.

According to a government report titled 'Chemical and Petrochemical Statistics at a Glance, 2024', of the 19 Mt of petrochemicals produced annually, 12.5 Mt, or 65.8 per cent, is used for polymer production. While plastic is widely seen as a 'wonder material', Singh thinks